

ORIGINAL RESEARCH ARTICLE

Incremental Prognostic Value of Subendocardial Myocardial Flow Reserve in Patients With Normal Perfusion

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BACKGROUND: Although the prognostic utility of positron emission tomography (PET) myocardial flow reserve (MFR) is well established, emerging data suggest that reduced subendocardial flows also predict adverse outcomes. However, the incremental value of subendocardial MFR (MFR_{SE}) beyond transmural MFR (MFR_{TM}) remains unclear.

METHODS: We studied patients in a multicenter PET registry with normal perfusion on stress/rest Rb-82 PET, excluding those with a previous history of coronary artery bypass surgery, heart transplantation, or left ventricular ejection fraction <40%. The optimal MFR_{SE} cutoff for predicting major adverse cardiovascular events (MACEs; death, myocardial infarction [MI], revascularization, or heart failure [HF] hospitalization) was determined using Youden's index. Patients were stratified into 3 groups: concordant-normal (MFR_{TM} ≥2.0; MFR_{SE} ≥2.1), discordant (low-MFR_{SE}, normal MFR_{TM}), and abnormal MFR_{TM}. Clinical outcomes were compared by MFR groups.

RESULTS: Among 6603 patients (normal N=4103; discordant N=885; abnormal N=1615) the mean age was 66.3±12.4 years, and 54% were women. Compared with the concordant-normal group, patients with discordant low-MFR_{SE} were older and more likely to have hypertension, diabetes, peripheral artery disease, and previous percutaneous coronary intervention. The median MFR_{TM} for normal, discordant, and abnormal groups were 2.86, 2.15, and 1.72, respectively. Over a median follow-up of 4.9 years, 1661 MACE events occurred. Discordant low-MFR_{SE} patients had a higher risk of MACE (hazard ratio [HR], 1.41; 95% CI, 1.22–1.64) and all-cause mortality (HR, 1.36; 95% CI, 1.14–1.61) compared with concordant-normal patients. The discordant group had an intermediate absolute risk of MACE, with an adjusted annualized event rate of 5.79% (95% CI, 5.10–6.49) compared with 3.99% (95% CI, 3.67–4.30; *P*<0.001) in the concordant-normal group and 8.35% (95% CI, 7.71–9.00; *P*<0.001) in the abnormal MFR_{TM} group.

CONCLUSIONS: Subendocardial MFR reveals clinically meaningful risk heterogeneity among patients with preserved transmural flow reserve, helping refine risk stratification beyond traditional PET metrics.

Key Words: coronary artery disease ■ microcirculation ■ myocardial perfusion imaging ■ positron-emission tomography

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Clinical Perspective

What Is New?

- Layer-specific positron emission tomography (PET) myocardial perfusion imaging revealed that 13.4% of patients had reduced subendocardial myocardial flow reserve (MFR_{SE}) despite preserved transmural MFR, identifying a previously unrecognized intermediate-risk subgroup.
- Patients with no evidence of flow-limiting coronary artery disease (CAD) and discordantly low MFR_{SE} had a 1.4-fold increased risk of major adverse cardiovascular events and all-cause mortality than those with normal transmural and subendocardial MFR.

What Are the Clinical Implications?

- Subendocardial flow assessment improves prognostic evaluation by revealing higher-risk patients not detected by conventional transmural flow quantification.
- Incorporating subendocardial flow metrics into PET interpretation provides a more detailed physiological profile that may guide diagnosis, risk assessment, and personalized treatment strategies in patients with nonobstructive CAD.

Nonstandard Abbreviations and Acronyms

CMD	coronary microvascular disease
MACE	major adverse cardiovascular event
MBF	myocardial blood flow
MFR	myocardial flow reserve
MFR_{SE}	subendocardial myocardial flow reserve
MFR_{TM}	transmural myocardial flow reserve
PET/CT	positron emission tomography/computed tomography

Coronary microvascular dysfunction (CMD) is a subtype of nonobstructive coronary artery disease (CAD) that is prevalent across a broad spectrum of cardiovascular (CV) risk factors and is associated with increased morbidity and death even in the absence of obstructive epicardial disease.^{1–5} The most validated and reproducible method for noninvasive assessment of CMD is positron emission tomography (PET) myocardial perfusion imaging (MPI), which quantifies myocardial flow reserve (MFR), the ratio of stress to rest myocardial blood flow (MBF).^{6–8} MFR reflects the integrated effects of diffuse atherosclerosis and microvascular dysfunction and serves as the current clinical standard for noninvasive assessment of coronary microvascular function.⁸

Conventional PET assessments derive MFR by averaging flow computations across the full myocardial wall

thickness (transmural MFR).⁹ Yet the subendocardium is particularly vulnerable to ischemia because of its higher metabolic demand and greater susceptibility to reductions in perfusion pressure.¹⁰ As a result, subendocardial flow abnormalities may precede obvious reductions in transmural MFR but remain undetected when absolute flow is averaged across myocardial layers.

Advances in PET imaging technology now allow for reliable layer-specific quantification of MFR. Although reduced subendocardial MFR has been associated with adverse CV outcomes,¹¹ its prognostic value among patients with preserved transmural MFR (those typically considered lower risk) remains unclear. In this study of patients with normal perfusion, we evaluated whether reduced subendocardial MFR identifies patients at increased risk for major adverse events despite preserved transmural MFR.

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.



Study Population

The study cohort was derived from a multicenter PET registry¹² and included consecutive patients who underwent rest–stress rubidium-82 PET between 2007 and 2024 with normal perfusion, defined as a stress total perfusion deficit (TPD) $\leq 5\%$ (equivalent to a summed stress score ≤ 3).¹³ Subendocardial flow data were available at 3 academic medical centers: Brigham and Women's Hospital (Boston, MA); Cedars-Sinai Medical Center (Los Angeles, CA); and Intermountain Medical Center (Salt Lake City, UT). Patients were excluded if they were <18 years of age, had a history of coronary artery bypass grafting, orthotopic heart transplantation, or other cardiac surgery, had a left ventricular (LV) ejection fraction $<40\%$, underwent nonvasodilator stress testing, lacked sufficient quality dynamic or gated series for image processing, or had no available clinical data (Figure 1). Clinical data, including demographics, past medical history, and indication for PET, were collected at the time of imaging by each participating site. The study was approved by the institutional review boards at each site, with central approval granted by the Cedars-Sinai Medical Center institutional review board. The study complied with the Declaration of Helsinki, and sites either obtained written informed consent or a waiver of consent for the use of the deidentified data.

Myocardial Perfusion PET/CT Imaging

PET rest and stress MPI was performed in accordance with clinical guidelines and the protocol of each study site.⁹ Patients were imaged with rubidium-82, and vasodilator stress was performed with regadenoson ($>90\%$), adenosine, or dipyridamole. A low-dose noncontrast computed tomography scan was used for attenuation correction. All images were centrally processed by the Cedars Sinai Core Laboratory in batch mode with dedicated software (QPET; Cedars-Sinai Medical Center, Los Angeles, CA).¹⁴ Reconstructed images were generated using vendor-recommended iterative reconstruction algorithms

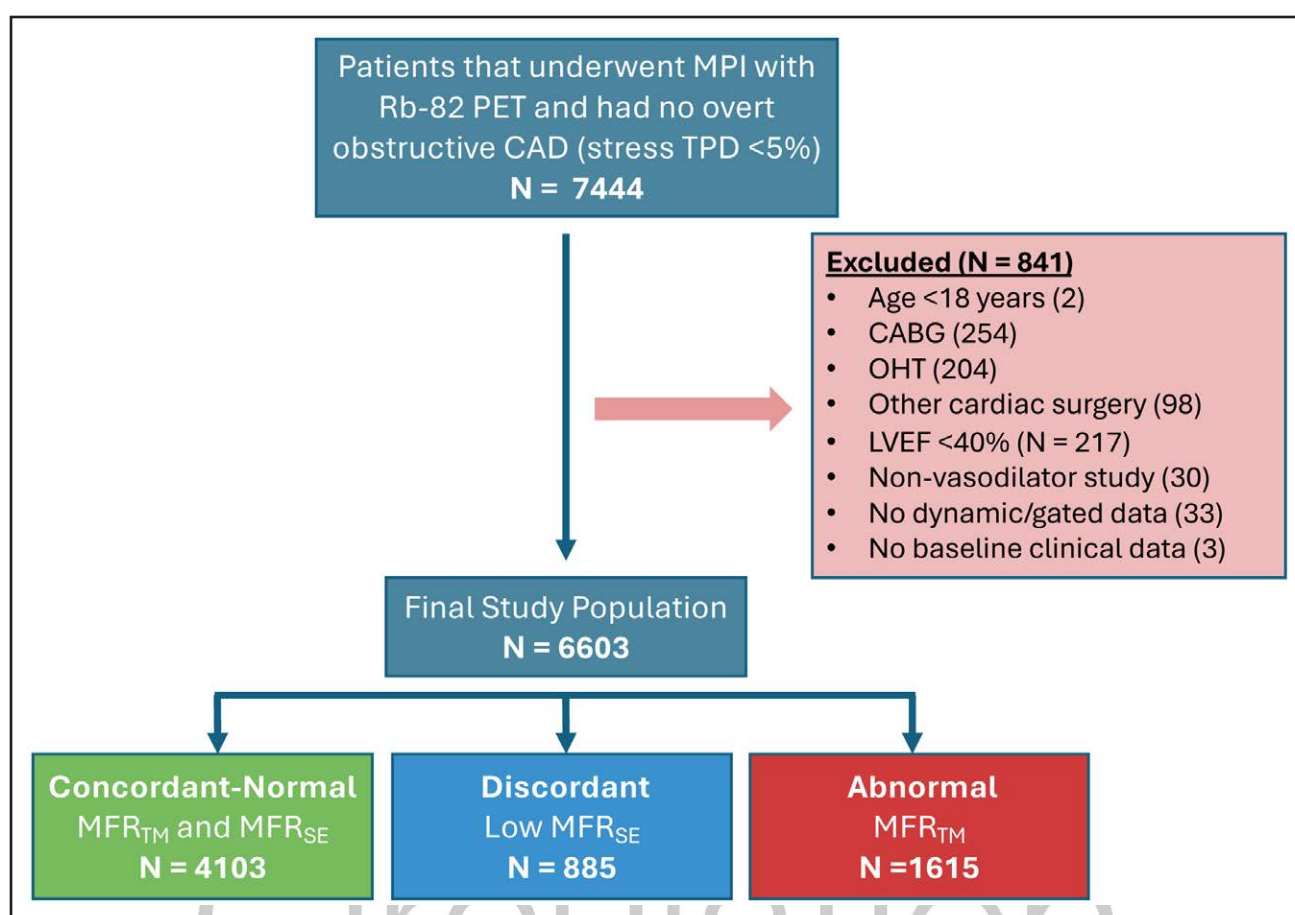


Figure 1. Study population flow diagram.

Final cohort selection (N=6603) and classification by transmural MFR (MFR_{TM}) and subendocardial MFR (MFR_{SE}). CABG indicates coronary artery bypass grafting; CAD, coronary artery disease; LVEF, left ventricular ejection fraction; MFR, myocardial flow reserve; MPI, myocardial perfusion imaging; OHT, orthotopic heart transplant; PET, positron emission tomography; and TPD, total perfusion deficit.

optimized for each scanner and site. Stress and rest relative perfusion were quantified using total perfusion deficit (TPD) for the entire left ventricle. Abnormal stress TPD was defined as $\geq 5\%$.¹³ LV ejection fraction (LVEF) reserve was calculated as the difference between stress and rest gated LVEF. LV mass was calculated from the gated images as previously described.¹⁵

Quantification of Myocardial Blood Flow and Flow Reserve

Myocardial blood flow and flow reserve maps were generated from the dynamic image series with the dedicated software using a one-compartment kinetic model (QPET; Cedars-Sinai Medical Center, Los Angeles, CA).¹⁶ To minimize partial volume effects, the blood-to-myocardium spillover fraction was estimated as a model parameter. Average MBF and MFR values were quantified for the entire left ventricle, each coronary vascular territory, and for each of the 17 myocardial segments in accordance with American Heart Association (AHA) standardized segmentation model.¹⁷ MFR was calculated as the ratio of stress MBF/rest MBF at each polar sample. A research version of QPET was used for layer-specific segmentation and blood flow quantification, as previously described.¹¹ The outer epicardial and inner endocardial contour surfaces of the myocardium were outlined. A midsurface was automatically created by standard geometric sampling that acted

as a boundary for the subendocardial counts. The subendocardial counts were obtained from the part of the myocardium that lies between the inner (endocardial) surface and the midsurface, resulting in segmentation of the LV wall into 2 layers of equal thickness. Automatic motion correction with frame-by-frame manual adjustment was applied to all dynamic studies.¹⁸ Distribution of quantified transmural, subendocardial, subepicardial flows and flow reserve are shown in Figure S1. Reduced transmural MFR (MFR_{TM}) was defined as <2.0 .⁹ As there are no established reference cutoffs for subendocardial MFR (MFR_{SE}) the threshold was selected based on optimal discrimination of major adverse cardiovascular event (MACE) risk in the study population. Patients were grouped into 3 categories: concordant-normal=normal MFR_{TM} and MFR_{SE}; discordant low MFR_{SE}=normal MFR_{TM} but reduced MFR_{SE}; and abnormal MFR_{TM}.

Quantification of Coronary Artery Calcium

Per-vessel coronary artery calcium (CAC) assessment using automated deep learning (DL) was performed on the attenuation-corrected CT images as previously described.¹⁹ Briefly, this model is comprised of 2 DL networks. The first DL network segmented the vessel-specific CAC and the second performed an ovoid segmentation to encompass the cardiac outline. A convolutional long short-term model with 3 slices was provided

to each network to replicate the multislice assessment performed during manual image analysis.

Clinical Outcomes

The primary outcome was time to first MACE, defined as a composite of all-cause death, nonfatal myocardial infarction (MI), heart failure (HF) hospitalization, or late coronary revascularization (≥ 90 days after PET scan). Secondary outcome was all-cause mortality. Clinical events were ascertained through electronic medical record review and adjudicated by physicians at each participating site. Deaths were verified using the National Death Index. MI was adjudicated in accordance with the Fourth Universal Definition of MI.²⁰ HF hospitalization was defined when a patient was discharged with a primary hospitalization diagnosis of HF or based on clinical presentation consistent with new or worsening HF. For each patient, follow-up duration was calculated from the index PET scan to the date of the first clinical event or the date of the last verified vital status at each site. The period of ascertainment for this analysis concluded with the latest site-specific adjudication update in December 2024.

Statistical Analysis

Categorical variables are reported as frequencies with percentages (%). Continuous variables are presented as mean $\pm$ standard deviation (SD) or median with 25th and 75th percentiles (Q1, Q3) where appropriate. Baseline characteristics and PET-derived variables were compared using the χ^2 test for categorical variables. Continuous variables were compared using ANOVA for normally distributed data and the Kruskal-Wallis test for non-normally distributed data. For determination of the optimal MFR_{SE} threshold for predicting MACE, the cohort was randomly split into derivation and validation sets (1:1), ensuring equal distribution of events. The optimal cutoff in the derivation set was determined using Youden's index. Kaplan-Meier curves were generated to visualize event-free survival for MACE and all-cause mortality. Cox proportional hazards models were used to estimate unadjusted and multivariable adjusted hazard ratios (HRs) for MACE and all-cause mortality. The proportional hazards assumption was verified using Schoenfeld residuals. Unadjusted and multivariable adjusted absolute event rates were estimated using Poisson regression with marginal standardization to the overall cohort. A subanalysis stratified Cox and Poisson models by sex. For the secondary composite endpoint of CV death or MI, cumulative incidence functions were used to estimate event probabilities with non-CV death treated as a competing risk. Subdistribution HRs were estimated using the Fine and Gray model. The multivariable models included adjustment for age, sex, diabetes, hypertension, hyperlipidemia, previous MI, previous percutaneous coronary intervention (PCI), body mass index, LV mass index, LVEF, and CAC. All tests were 2-sided, and a P value <0.05 was considered statistically significant. Statistical analysis was performed with the use of Stata/BE 17.0 (Statacorp, College Station, TX).

RESULTS

Study Population

The optimal threshold for subendocardial MFR was determined to be 2.1 based on split-sample derivation and

validation (Table S1). Using this threshold, the final study cohort ($N=6603$) included 4103 (62.1%) concordant-normal, 885 (13.4%) discordant low-MFR_{SE}, and 1615 (24.5%) abnormal MFR_{TM} (Figure 1). In the abnormal MFR_{TM} group, MFR_{SE} was also reduced in nearly all cases; only 20 (1.2%) had preserved MFR_{SE} >2.1 . Baseline patient characteristics stratified by MFR groups are presented in Table 1. The mean age was 66.3 ± 12.4 years, 45.9% were men, and the majority were White (79.6%). Cardiometabolic comorbidities were common, including hypertension (78.9%), obesity (41.5%), and diabetes (28.0%). Compared with the concordant-normal group, patients with discordant low-MFR_{SE} were older and more likely to have hypertension, diabetes, peripheral artery disease, and previous PCI. The abnormal MFR_{TM} group had the greatest overall burden of comorbidities. Chest pain was the most common indication for PET MPI across all groups ($>50\%$), but dyspnea and preoperative evaluation were relatively more common referral reasons in the discordant low-MFR_{SE} and abnormal MFR_{TM} groups (Table 1).

PET Characteristics



Figure S1 demonstrates the distribution of MBF and MFR across the transmural, subendocardial, and subepicardial layers for the overall cohort. Subendocardial MBF was higher than subepicardial MBF at both rest and stress ($P<0.0001$ for both), and the relative difference between these layers narrowed during stress. Subendocardial MFR was lower than subepicardial MFR (2.36 versus 2.72; $P<0.0001$), with transmural MFR (2.51 [IQR, 2.01–3.01]) falling between those values. Figure 2 illustrates the distribution of regional MFRs stratified by MFR group, with the relative separation between subendocardial and epicardial MFR most apparent in the discordant group. Although both the concordant and discordant groups had preserved (≥ 2.0) transmural MFR, values were significantly lower in the discordant group (2.15 [2.06, 2.28] versus 2.85 [2.51, 3.32]; $P<0.001$).

PET-derived flow and functional parameters are summarized in Table 2. Across all myocardial layers (subendocardial, transmural, and subepicardial), there was a consistent stepwise decline in stress MBF and MFR, and a parallel increase in rest MBF, from the concordant-normal group to the discordant low-MFR_{SE} and abnormal MFR_{TM} groups. LV mass index differed statistically significantly across groups, although the absolute differences were small and showed no consistent directional pattern. Transient ischemic dilation (TID) was more frequently observed in the discordant low-MFR_{SE} group, and negative LVEF reserve was more common in the discordant low-MFR_{SE} (18.2%) and abnormal MFR_{TM} groups (25.0%) when compared with the concordant-normal group (12.4%). CAC burden also increased stepwise across MFR groups, mirroring

Table 1. Baseline Patient Characteristics by Subendocardial-Transmural MFR Groups

	Total	MFR _{TM} ≥2		MFR _{TM} <2	P value
		Concordant-normal (MFR _{SE} >2.1)	Discordant low-MFR _{SE} (MFR _{SE} <2.1)	Abnormal MFR _{TM}	
	N=6603	N=4103	N=885	N=1615	
Demographics					
Age, y	66.3±12.4	64.3±11.9	68.8±11.7	70.1±12.8	<0.001
Male	3033 (45.9%)	1980 (48.3%)	391 (44.2%)	662 (41.0%)	<0.001
White	5253 (79.6%)	3334 (81.3%)	697 (78.8%)	1222 (75.7%)	<0.001
Black	665 (10.1%)	384 (9.4%)	97 (11.0%)	184 (11.4%)	0.045
Medical history					
BMI ≥30 kg/m ²	2741 (41.5%)	1744 (42.5%)	340 (38.4%)	657 (40.7%)	0.060
Hyperlipidemia	4654 (71.2%)	2904 (71.4%)	625 (71.6%)	1125 (70.7%)	0.83
Hypertension	5155 (78.9%)	3080 (75.7%)	707 (80.1%)	1368 (85.9%)	<0.001
Diabetes	1845 (28.0%)	940 (22.9%)	259 (29.3%)	646 (40.0%)	<0.001
Smoking	1463 (22.4%)	1014 (24.9%)	164 (18.5%)	285 (17.9%)	<0.001
History of MI	581 (8.8%)	334 (8.1%)	67 (7.6%)	180 (11.1%)	<0.001
History of PCI	725 (11.0%)	391 (9.5%)	121 (13.7%)	213 (13.2%)	<0.001
PAD	288 (4.4%)	122 (3.0%)	44 (5.0%)	122 (7.6%)	<0.001
PET indication					
Chest pain	4325 (65.5%)	2895 (70.6%)	561 (63.4%)	869 (53.8%)	<0.001
Dyspnea	226 (3.4%)	105 (2.6%)	43 (4.9%)	78 (4.8%)	<0.001
Abnormal study	467 (7.1%)	309 (7.5%)	54 (6.1%)	104 (6.4%)	0.17
Preoperative evaluation	507 (7.7%)	217 (5.3%)	79 (8.9%)	211 (13.1%)	<0.001
Other	1078 (16.3%)	577 (14.1%)	148 (16.7%)	353 (21.9%)	<0.001

Values are mean±standard deviation or number (%).
BMI indicates body mass index; MI, myocardial infarction; PAD, peripheral arterial disease; PCI, percutaneous coronary intervention;
and PET, positron emission tomography.

the directional pattern observed in PET-derived physiological measures (Table 2). Similar distributions of subendocardial, transmural, and subepicardial MFR were observed across groups when stratified by sex (Table S2).

Clinical Outcomes

Follow-up data were available for 6584 of 6603 patients (99.7%), with a median follow-up of 4.95 years (Q1, Q3: 4.19, 7.11). The primary outcome of composite MACE occurred in 1661 individuals (24.9% of the cohort), consisting of 1223 deaths, 218 MIs, 311 HF admissions, and 284 late revascularizations (Table S3). Among patients who experienced a MACE, death was the first event for 60.9%, followed by HF admission (16.8%), MI (11.8%), and late revascularization (10.6%).

Time-to-event outcomes differed significantly across the MFR groups (Figure 3). Unadjusted Cox regression results are summarized in Table 3. In multivariable analysis, patients with discordant low-MFR_{SE} had a higher risk of MACE (HR, 1.41; 95% CI, 1.22–1.63) and all-cause mortality (HR, 1.36; 95% CI, 1.14–1.61) compared with concordant-normal patients. Compared with the discor-

dant group, the abnormal MFR_{TM} group had a higher risk of MACE (HR, 1.46; 95% CI, 1.26–1.68) and all-cause mortality (HR, 1.54; 95% CI, 1.31–1.82; Table 3). These results remained unchanged after further adjusting for the year of the PET scan.

Absolute event rates are summarized in Table 4. The discordant low-MFR_{SE} group had intermediate absolute risk of MACE, with an adjusted annualized event rate (AER) of 5.79% (95% CI, 5.10–6.49), compared with 3.99% (95% CI, 3.67–4.30) in the concordant-normal group and 8.35% (95% CI, 7.71–9.00) in the abnormal MFR_{TM} group. A similar stepwise increase in mortality risk was observed, with adjusted AERs of 2.61% (95% CI, 2.36–2.86) in the concordant-normal group, 3.73% (95% CI, 3.21–4.27) in the discordant low-MFR_{SE} group, and 5.69% (95% CI, 5.19–6.18) in the abnormal MFR_{TM} group.

The association between MFR groups and time to MACE was consistent across sexes. The discordant group was associated with an intermediate risk of adverse events in both men and women, although the risk for all-cause mortality in women with discordant versus normal MFR was no longer significant after multivariable adjustment (Table 3; Table S4).

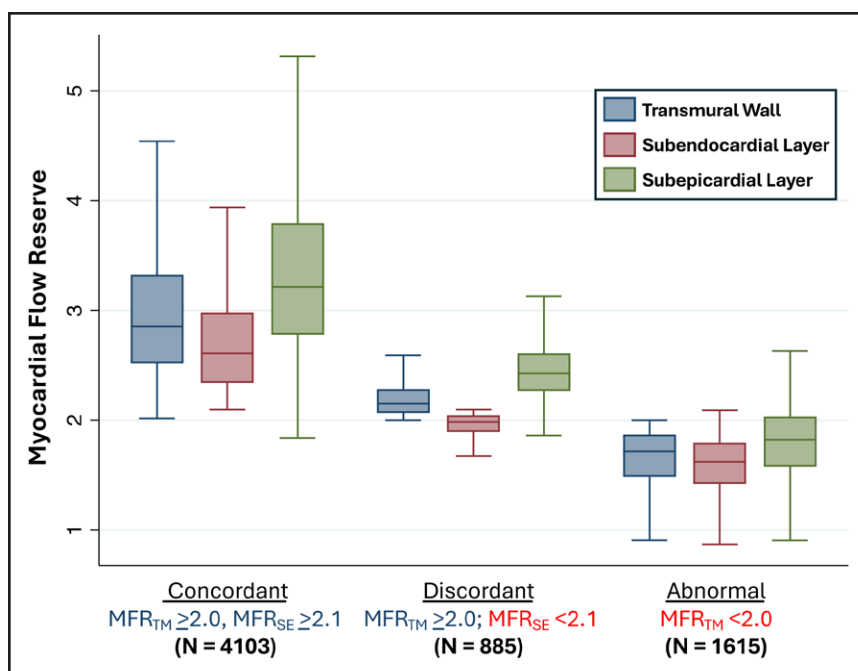


Figure 2. Distribution of transmural, subendocardial, and subepicardial MFR by MFR groups.

Each box represents the interquartile range (IQR), with the line inside indicating the median and whiskers extending to 1.5×IQR. MFR indicates myocardial flow reserve; MFR_{SE} subendocardial MFR; and MFR_{TM} transmural MFR.

In a secondary analysis of a CV-specific composite MACE (CV death, MI, and HF admission), there were 705 total events, including 278 CV deaths. Discordant low-MFR_{SE} was associated with a higher risk of MACE com-

pared with the concordant-normal group (HR, 1.53; 95% CI, 1.22–1.91; $P < 0.001$). Compared with the discordant group, the abnormal MFR_{TM} group had a higher risk of MACE (HR, 1.46, 95% CI, 1.17–1.82; $P = 0.001$).

Table 2. Stress PET Results by MFR Groups

	Total	MFR _{TM} ≥ 2		MFR _{TM} < 2	P value
		Concordant (MFR _{SE} ≥ 2.1)	Discordant (MFR _{SE} < 2.1)	Abnormal MFR _{TM}	
	N=6603	N=4103	N=885	N=1615	
Subendocardial MFR	2.28 (1.90, 2.75)	2.61 (2.34, 2.98)	1.98 (1.90, 2.04)	1.63 (1.43, 1.80)	<0.001
Subendocardial rest MBF	1.32 (1.09, 1.56)	1.22 (1.01, 1.43)	1.40 (1.20, 1.62)	1.54 (1.33, 1.82)	<0.001
Subendocardial stress MBF	2.99 (2.46, 3.53)	3.23 (2.77, 3.69)	2.70 (2.29, 3.17)	2.43 (1.99, 2.98)	<0.001
Transmural MFR	2.46 (2.01, 3.01)	2.85 (2.51, 3.32)	2.15 (2.06, 2.28)	1.72 (1.48, 1.87)	<0.001
Transmural rest MBF	1.11 (0.88, 1.37)	1.00 (0.81, 1.24)	1.15 (0.97, 1.39)	1.37 (1.15, 1.64)	<0.001
Transmural stress MBF	2.71 (2.21, 3.25)	2.92 (2.48, 3.40)	2.51 (2.10, 2.98)	2.23 (1.87, 2.75)	<0.001
Subepicardial MFR	2.73 (2.18, 3.42)	3.21 (2.77, 3.79)	2.42 (2.26, 2.61)	1.82 (1.58, 2.03)	<0.001
Subepicardial rest MBF	0.86 (0.66, 1.103)	0.76 (0.60, 0.98)	0.90 (0.73, 1.10)	1.11 (0.89, 1.40)	<0.001
Subepicardial stress MBF	2.33 (1.89, 2.82)	2.51 (2.09, 2.97)	2.17 (1.79, 2.65)	1.93 (1.56, 2.39)	<0.001
Resting SBP, mm Hg	130 (117, 145)	127 (114, 141)	133 (121, 150)	138 (124, 155)	<0.001
Resting DBP, mm Hg	66 (57, 75)	66 (57, 75)	67 (57, 76)	67 (58, 76)	0.081
Resting heart rate, bpm	69 (61, 78)	68 (60, 76)	68 (61, 77)	72 (65, 82)	<0.001
LV mass index, g/m ²	74.6 (66.1, 84.1)	74.2 (66.1, 82.9)	73.9 (64.8, 83.3)	76.3 (66.8, 88.2)	<0.001
Resting LVEF, %	69.7 (62.9, 76.3)	70.0 (63.5–76.5)	69.4 (62.3, 76.2)	69.2 (61.7, 76.1)	0.005
LVEF reserve <0	1074 (16.3%)	509 (12.4%)	161 (18.2%)	404 (25.0%)	<0.001
TID >1.1	1440 (21.8%)	844 (20.6%)	228 (25.8%)	368 (22.8%)	0.002
CAC score, AU	39.9 (0, 388.8)	21.7 (0, 271.0)	78.5 (0, 551.4)	106.3 (0, 624.7)	<0.001

Values are median (Q1, Q3) or number (%).

AU indicates Agatston Units; CAC, coronary artery calcium; DBP, diastolic blood pressure; LV, left ventricular; LVEF, left ventricular ejection fraction; MBF, myocardial blood flow; MFR, myocardial flow reserve; MFR_{SE} subendocardial MFR; MFR_{TM} transmural MFR; SBP, systolic blood pressure; and TID, transient ischemic dilation.

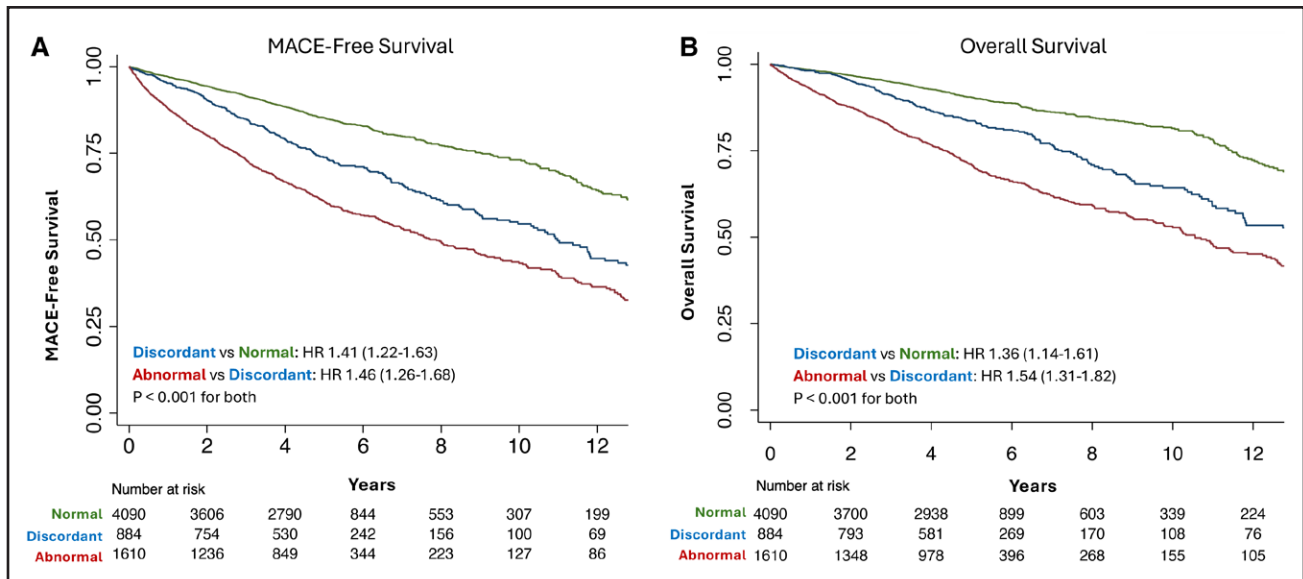


Figure 3. Event-free survival by transmural and subendocardial MFR groups.

Kaplan-Meier curves showing MACE-free survival (A) and overall survival (B) stratified by MFR group (normal, $MFR_{TM} \geq 2.0$ and $MFR_{SE} \geq 2.1$; discordant, $MFR_{TM} \geq 2.0$ and $MFR_{SE} < 2.1$; abnormal, $MFR_{TM} < 2.0$). Adjusted hazard ratios (HRs) from Cox proportional hazards models are shown for pairwise comparisons. Models adjusted for age, sex, diabetes, hypertension, hyperlipidemia, previous myocardial infarction, previous percutaneous coronary intervention, peripheral artery disease, body mass index, left ventricular mass index, left ventricular ejection fraction, and coronary artery calcium score. MACE indicates major adverse cardiovascular event (all-cause mortality, myocardial infarction, heart failure admission, or late revascularization). MFR, myocardial flow reserve; MFR_{SE} , subendocardial MFR; and MFR_{TM} , transmural MFR.

DISCUSSION

In this large, multicenter cohort of patients without evidence of flow-limiting CAD, layer-specific flow analysis identified a sizeable subgroup (13.4%) with isolated

reduction in MFR_{SE} despite normal MFR_{TM} . Compared with concordant-normal patients, the discordant phenotype had a higher burden of cardiometabolic risk factors, including hypertension and diabetes, and more CAC. Whereas T1D and poor LVEF reserve were more common

Table 3. Unadjusted and Adjusted Hazard Ratios for MACE and All-Cause Mortality by MFR Group

	N events	Comparison	Unadjusted HR (95% CI)	P value	Adjusted HR(95% CI)	P value
All (N=6584)						
MACE	1661	Discordant vs normal	1.83 (1.59–2.11)	<0.001	1.41 (1.22–1.63)	<0.001
		Abnormal vs discordant	1.60 (1.39–1.84)	<0.001	1.46 (1.26–1.68)	<0.001
ACM	1200	Discordant vs normal	1.80 (1.52–2.13)	<0.001	1.36 (1.14–1.61)	0.001
		Abnormal vs discordant	1.75 (1.40–2.06)	<0.001	1.54 (1.31–1.82)	<0.001
Women (N=3560)						
MACE	863	Discordant vs normal	1.72 (1.41–2.09)	<0.001	1.36 (1.12–1.67)	0.002
		Abnormal vs discordant	1.72 (1.42–2.09)	<0.001	1.39 (1.15–1.70)	0.001
ACM	653	Discordant vs normal	1.56 (1.24–1.97)	<0.001	1.20 (0.94–1.51)	0.137
		Abnormal vs discordant	1.83 (1.46–2.29)	<0.001	1.48 (1.17–1.86)	0.001
Men (N=3024)						
MACE	783	Discordant vs normal	2.01 (1.65–2.44)	<0.001	1.52 (1.24–1.87)	<0.001
		Abnormal vs discordant	1.50 (1.23–1.83)	<0.001	1.48 (1.20–1.83)	<0.001
ACM	547	Discordant vs normal	2.14 (1.68–2.72)	<0.001	1.60 (1.24–2.05)	<0.001
		Abnormal vs discordant	1.70 (1.34–2.15)	<0.001	1.62 (1.26–2.07)	<0.001

MFR groups: normal= $MFR_{TM} \geq 2$ and $MFR_{SE} \geq 2.1$; discordant= $MFR_{TM} \geq 2$ and $MFR_{SE} < 2.1$, and abnormal= $MFR_{TM} < 2$. Values reflect unadjusted and adjusted hazard ratios with 95% CIs from Cox proportional hazards models. Adjusted models account for age, sex, diabetes, hypertension, hyperlipidemia, previous myocardial infarction, previous percutaneous coronary intervention, body mass index, left ventricular mass index, left ventricular ejection fraction, and coronary artery calcium.

ACM indicates all-cause mortality; HR, hazard ratio; MACE, major adverse cardiovascular event; MFR, myocardial flow reserve; MFR_{SE} , subendocardial MFR; and MFR_{TM} , transmural MFR.

Table 4. Annualized Event Rates by MFR Groups

Group	Unadjusted AER (%/year)	P value	Adjusted AER (%/year)	P value
MACE				
Concordant-normal	3.27 (3.02–3.51)	Ref	3.99 (3.67–4.30)	Ref
Discordant low MFR _{SE}	6.00 (5.29–6.70)	<0.001	5.79 (5.10–6.49)	<0.001
Abnormal MFR _{TM}	9.58 (8.87–10.29)	<0.001*	8.35 (7.71–9.00)	<0.001*
ACM				
Concordant-normal	2.16 (1.96–2.35)	Ref	2.61 (2.36–2.86)	Ref
Discordant low MFR _{SE}	3.97 (3.41–4.52)	<0.001	3.73 (3.21–4.27)	<0.001
Abnormal MFR _{TM}	6.84 (6.28–7.41)	<0.001*	5.69 (5.19–6.18)	<0.001*

*Comparison with discordant low MFR_{SE}. Adjusted rates account for age, sex, diabetes, hypertension, hyperlipidemia, previous myocardial infarction, previous percutaneous coronary intervention, peripheral artery disease, body mass index, left ventricular mass index, left ventricular ejection fraction, and coronary artery calcium score.

ACM indicates all-cause mortality; AER, annualized event rate; MACE, major adverse cardiovascular event; MFR, myocardial flow reserve; MFR_{SE}, subendocardial MFR; and MFR_{TM}, transmural MFR.

in these patients compared with concordant-normal, the majority (>70%) lacked these findings, underscoring the limitations of classic markers of subendocardial ischemia to identify subendocardial flow abnormalities. Importantly, integrating subendocardial assessment reclassified 17.8% of patients (885 of the 4988 patients with normal MFR_{TM}), revealing a subgroup with a 1.4-fold higher adjusted risk of MACE and all-cause mortality. These findings demonstrate that layer-specific perfusion analysis reveals prognostically meaningful heterogeneity among patients with preserved MFR_{TM}.

The ability of subendocardial flow impairment to unmask risk despite preserved MFR_{TM} is consistent with the distinct physiological stresses faced by this layer. The subendocardium experiences the highest systolic wall stress, has thinner arteriolar walls that increase susceptibility to compression, and faces disproportionately high metabolic demand.¹⁰ Even modest elevations in LV pressure or heart rate can exacerbate subendocardial underperfusion because of transmural flow redistribution.¹⁰ This vulnerability of the subendocardium has been demonstrated across a spectrum of disease states. In pressure-overload conditions such as aortic stenosis and hypertrophic cardiomyopathy, PET-derived subendocardial flows fall earlier and more steeply than subepicardial flow in the disease process and track closely with markers of increased workload and wall stress.^{21,22}

Notably, subendocardial flow abnormalities are not limited to patients with significant structural remodeling and may therefore signal early microvascular dysfunction. In a study of patients with hypertension and no flow-limiting CAD (most of whom had normal to mild increases in LV wall thickness), lower MFR_{SE} was associated with higher levels of N-terminal pro-B-type natriuretic peptide and high-sensitivity troponin T, independent of CV risk factors or atherosclerotic burden.¹¹ Although reduced subendocardial perfusion has been linked to increased LV mass,^{11,22} this relationship appears to be neither exclusive nor dominant. Rajappan et al. demon-

strated that in patients with aortic stenosis, reductions in subendocardial flow were more closely related to AS severity and diastolic abnormalities than to LV mass.²¹ In our cohort, only modest differences in indexed LV mass were observed across MFR groups, suggesting that it was not the primary driver of the discordance between subendocardial and transmural flows.

Although subendocardial vulnerability has been described physiologically, few studies have linked subendocardial flow abnormalities to hard clinical outcomes. In a single-center study of patients with hypertension and no perfusion defects, Xu et al found that lower subendocardial MFR independently predicted adverse events, including all-cause mortality and HF. In a large cohort of patients undergoing dipyridamole rubidium-82 PET, Gould et al found that relative subendocardial perfusion abnormalities were common, even among those without severely reduced coronary flow capacity.²³ The lower event rates they reported (9% over a median follow-up of 4.5 years) may reflect differences in referral population or in the methodological approach, as their subendocardial assessment was based on relative perfusion rather than absolute flow quantification. The present study, spanning >6000 patients across 3 institutions, is the first to demonstrate that absolute subendocardial flow quantification adds prognostic value beyond standard transmural MFR assessment, identifying a higher-risk subgroup that would otherwise be classified as low risk by the absence of perfusion abnormalities and preserved MFR_{TM} (>2.0). Furthermore, this added prognostic utility was observed in both men and women, underscoring its relevance across both sexes.

These findings come at a time when many patients continue to experience chest pain or dyspnea and remain at increased risk for adverse events, even in the absence of overt obstructive CAD.^{24,25} Although a 2-fold increase in transmural MBF is a common low-risk benchmark, our data suggest that this binary definition overlooks

significant risk heterogeneity.⁸ Because transmural MFR represents a full-thickness average, the preserved flow in the subepicardial layer can numerically mask significant impairment confined to the subendocardium, effectively diluting the diagnostic signal of early microvascular disease. The observation that MFR_{TM} values in the discordant group were clustered near the lower bound of the normal range, compared with the robustly normal values in the concordant group, reflects how preserved flow in other layers can numerically offset or mask significant impairment localized to the subendocardium. Consequently, a transmural average may fail to capture early microvascular dysfunction before it reaches a level of global severity. Subendocardial assessment, by capturing early or localized flow impairment, may help flag patients who remain at increased risk. Taken together, these findings highlight the gap between persistent clinical risk and standard flow evaluation and suggest that more comprehensive physiological assessment with PET MPI could better reflect underlying disease biology.

Additional studies are warranted to clarify the clinical significance of the Discordant phenotype. Future work should assess whether isolated reductions in subendocardial MFR contribute to symptoms, particularly among patients with normal transmural flow, to improve understanding of angina mechanisms in the absence of obstructive CAD. Integration of anatomic data, such as coronary CT angiography or invasive coronary angiography, is also needed to determine whether subendocardial flow assessment can further refine CAD phenotyping, including differentiation between obstructive and nonobstructive disease.

Limitations

This study has several limitations. First, as an observational retrospective study of a clinically referred cohort, generalizability may be limited, and residual confounding cannot be excluded despite adjustment for key covariates. Second, because subendocardial flow assessment is not yet in widespread clinical use, there are no established reference values. Larger studies in healthy reference populations are necessary to define normative distributions, which would enhance interpretability and support broader clinical application. Third, because of limitations in spatial resolution, subendocardial flow was approximated using the inner half of the LV wall, which likely includes central myocardial layers as well, and may lead to overestimation of MFR_{SE}. Fourth, although the robust event rate suggests that high data capture and all-cause mortality were independently verified, incomplete ascertainment of hospitalizations occurring outside of the participating systems remains possible. Fifth, as angiographic coronary anatomy was not available, we cannot definitively exclude the presence of multivessel CAD in patients with normal relative

perfusion. However, contemporary PET data suggest that balanced ischemia with multivessel epicardial disease is uncommon (<3%).²⁶ Finally, all patients underwent Rb-82 PET imaging; further research is needed to evaluate the added value of subendocardial flow assessment across other perfusion tracers, including ¹³N-ammonia and ¹⁸F-flurpiridaz.

Conclusions

Impaired subendocardial MFR identifies a higher-risk subgroup among symptomatic patients with preserved transmural MFR and normal perfusion. This finding reclassifies risk for nearly one in 5 patients considered low risk based on the widely applied clinical threshold of MFR>2.0, providing physiological granularity that enhances risk stratification.

ARTICLE INFORMATION

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Supplemental Material

Figure S1
Tables S1–S4

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